PHYS 209: Aspects of Modern Physics

Module 4: Qubits

Lecture 2: The Qubit as a Sensor

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Review

- A qubit state lives on the surface of a Bloch sphere

$$|\psi\rangle = \cos\frac{\theta}{2}|0\rangle + \sin\frac{\theta}{2}e^{i\phi}|1\rangle = \begin{bmatrix} \cos\frac{\theta}{2} \\ \sin\frac{\theta}{2}e^{i\phi} \end{bmatrix}$$

- Pauli Matrices act on qubits as operators

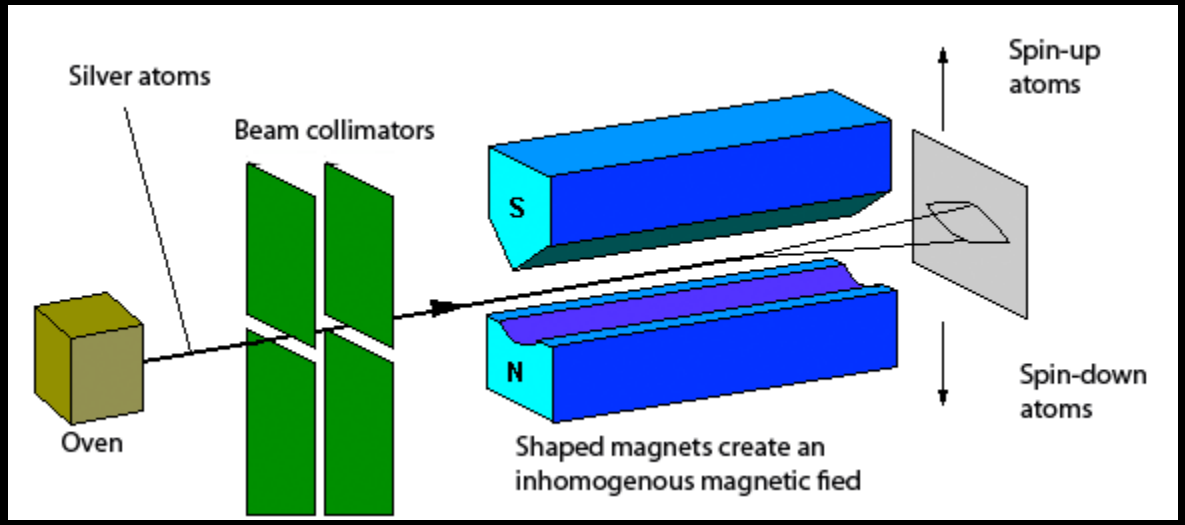
$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \qquad Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}, \qquad Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

  - These represent rotations of 180° around the corresponding axes.

Stern-Gerlach Experiment

- Silver atoms have 0 charge.
- Silver atoms have 1 valence electron.
- This experiment showed that electrons have an internal angular momentum called spin which is quantized and can take up only two values.
- Electrons have Spin ½.

$$S = 1/2$$
$$m_s = \pm 1/2$$



https://www.informationphilosopher.com/solutions/experiments/stern_gerlach/

- Electron spin states can be represented as

$\|1/2\rangle$	$\|\uparrow\rangle$	$\|0\rangle$
$\|-1/2\rangle$	$\|\downarrow\rangle$	$\|1\rangle$

Magnetic Moment

- Place a Spin ½ particle in static magnetic field $\vec{B} = B_0 \vec{e}_z$
- Classical energy of a magnetic dipole $\vec{\mu}$ in magnetic field $\vec{B}$ is

$$E = -\vec{\mu} \cdot \vec{B}$$

- $\vec{B}$ produces a torque on $\vec{\mu}$

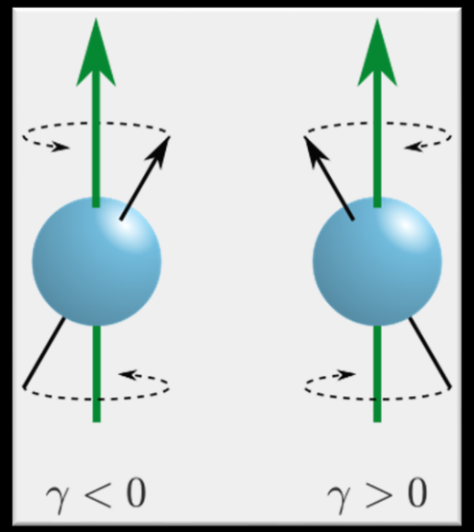
$$\vec{\tau} = \vec{\mu} \times \vec{B}$$

- The magnetic moment of a spin is

$$\vec{\mu} = \gamma \vec{S}$$

  - $\vec{S}$ is spin angular momentum
  - Gyromagnetic ratio γ relates a particle's magnetic moment to its spin angular momentum

Precession



$\gamma_{e^-} \approx -1.76 \times 10^{11}\ \mathrm{s^{-1}T^{-1}}$
$\gamma_p \approx 2.68 \times 10^{8}\ \mathrm{s^{-1}T^{-1}}$
$\gamma_n \approx -1.83 \times 10^{8}\ \mathrm{s^{-1}T^{-1}}$

Hamiltonian

- Energy: $E = -\vec{\mu} \cdot \vec{B}$
  - Magnetic Field: $\vec{B} = B_0 \vec{e}_z$
  - Magnetic Moment: $\vec{\mu} = \gamma \vec{S}$
- In Quantum Mechanics, the spin angular momentum becomes an operator

$$\hat{\vec{S}} = \frac{\hbar}{2}\hat{\vec{\sigma}} = \frac{\hbar}{2}\left(\hat{X}\vec{e}_x + \hat{Y}\vec{e}_y + \hat{Z}\vec{e}_z\right)$$

- And the energy operator is called Hamiltonian

$$\hat{H} = -\gamma \frac{\hbar}{2}\vec{\sigma} \cdot B_0 \vec{e}_z = -\frac{\hbar}{2}\gamma B_0 \hat{Z} = -\frac{1}{2}\hbar\omega_0 \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

Larmor Frequency
$\omega_0 = \gamma B_0$

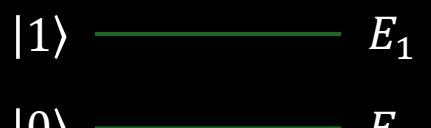
Eigenenergies

$$\hat{H} = -\frac{1}{2}\hbar\omega_0 \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

- Remember Schrodinger's Equation: $\hat{H}|\psi\rangle = E|\psi\rangle$
- Remember

$$\hat{Z}|0\rangle = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}\begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} = |0\rangle,$$

$$\hat{Z}|1\rangle = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}\begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \end{bmatrix} = -|1\rangle$$



- Applying the Hamiltonian to the computational basis states

$$\hat{H}|0\rangle = -\frac{1}{2}\hbar\omega_0|0\rangle, \qquad \hat{H}|1\rangle = \frac{1}{2}\hbar\omega_0|1\rangle$$

- The eigenenergies are

$$E_0 = -\frac{1}{2}\hbar\omega_0, \qquad E_1 = \frac{1}{2}\hbar\omega_0$$

Time Evolution in Quantum Mechanics

- Time-Independent Schrodinger Equation

$$\hat{H}|\psi\rangle = E|\psi\rangle$$

- Time-Dependent Schrodinger Equation

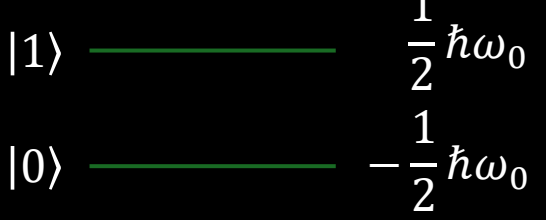
$$\hat{H}|\psi(t)\rangle = i\hbar\frac{d}{dt}|\psi(t)\rangle$$

- The general solution can be written as

$$|\psi(t)\rangle = e^{-i\hat{H}t/\hbar}|\psi(0)\rangle$$

  - The Hamiltonian determines how fast and in what way the state changes in time.

Time Evolution of Spin State



- Suppose the initial state is the superposition state

$$|\psi(0)\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

- Under the influence of a magnetic field characterized by the Hamiltonian $\hat{H} = -\frac{1}{2}\hbar\omega_0 \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$, the state evolves as

$$|\psi(t)\rangle = \exp\left[-\frac{i}{\hbar}\left(-\frac{1}{2}\hbar\omega_0\hat{Z}\right)t\right]\frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

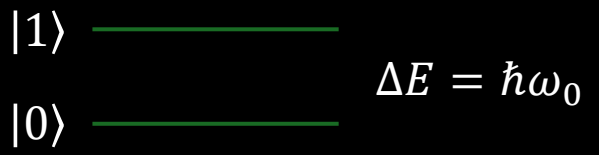
$$= \frac{\exp\left[i\frac{1}{2}\omega_0 t\right]|0\rangle + \exp\left[-i\frac{1}{2}\omega_0 t\right]|1\rangle}{\sqrt{2}}$$

$$\hat{Z}|0\rangle = |0\rangle$$
$$\hat{Z}|1\rangle = -|1\rangle$$

- Getting rid of the global phase,

$$|\psi(t)\rangle = \frac{|0\rangle + e^{-i\omega_0 t}|1\rangle}{\sqrt{2}}$$

Larmor Precession

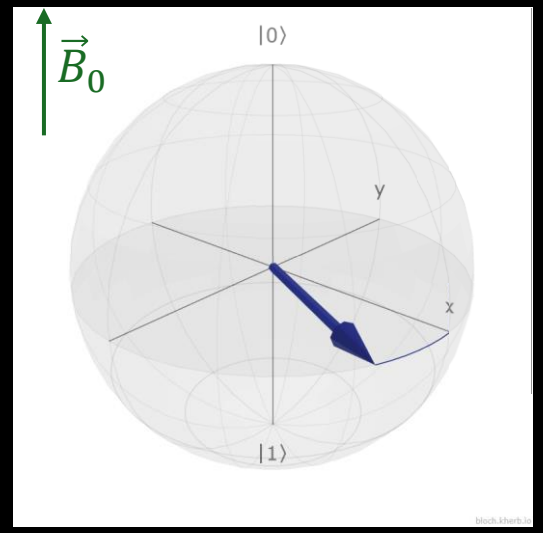


- The final spin state is

$$|\psi(t)\rangle = \frac{|0\rangle + e^{-i\omega_0 t}|1\rangle}{\sqrt{2}}$$

Larmor Frequency
$\omega_0 = \gamma B_0$

- The accumulated phase is $\phi = \omega_0 t = \gamma B_0 t$.
- The state rotates around z-axis with frequency ω_0.
- This is called Larmor precession.
  - The larger the applied magnetic field B_0, the larger the energy splitting $\Delta E = \hbar\omega_0$, and faster the precession.



Driving the Spin State

- Static field $\vec{B}_0 = B_0 \vec{e}_z$ forces spin to precess at $\omega_0 = \gamma B_0$.
- Apply second magnetic field:

$$\vec{B}_1(t) = B_1 \cos(\omega t)\, \vec{e}_x$$

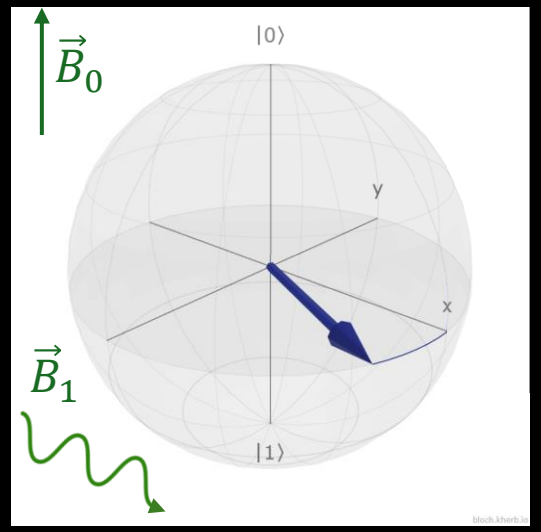
- Classical Energy:

$$U = -\vec{\mu} \cdot \vec{B}_1 = -\mu_x B_1 \cos(\omega t)$$

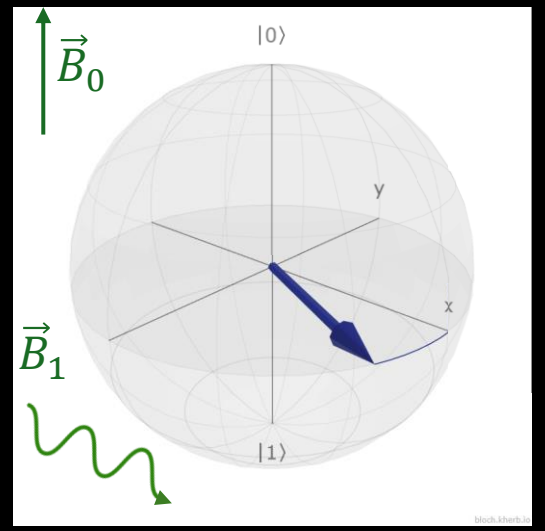
- Hamiltonian:

$$H = -\gamma \frac{\hbar}{2} \hat{X} B_1 \cos(\omega t) = -\frac{1}{2} \hbar \Omega \cos(\omega t)\, \hat{X}$$

  - Rabi frequency: $\Omega = \gamma B_1$
  - This is a time-dependent Hamiltonian!



Driving the Spin State



$$\vec{B} = \vec{B}_0 + \vec{B}_1(t) = B_0\vec{e}_z + B_1\cos(\omega t)\,\vec{e}_x$$

$$H = -\frac{1}{2}\hbar\omega_0\hat{Z} - \frac{1}{2}\hbar\Omega\cos(\omega t)\,\hat{X}$$

$$= -\frac{1}{2}\hbar\omega_0\begin{bmatrix}1 & 0\\ 0 & -1\end{bmatrix} - \frac{1}{2}\hbar\Omega\cos(\omega t)\begin{bmatrix}0 & 1\\ 1 & 0\end{bmatrix}$$

$$= -\frac{1}{2}\hbar\begin{bmatrix}\omega_0 & \Omega\cos(\omega t)\\ \Omega\cos(\omega t) & -\omega_0\end{bmatrix}$$

- The time-dependent Hamiltonian requires advanced techniques to solve.
- It leads to Generalized Rabi Formula:

$$P_{|1\rangle}(t) = \frac{\Omega^2}{\Omega^2 + (\omega - \omega_0)^2}\sin^2\left(\frac{\sqrt{\Omega^2 + (\omega - \omega_0)^2}}{2}t\right)$$

Rabi Oscillations

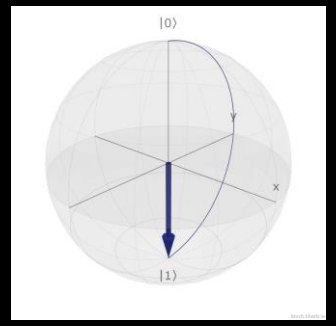
$$P_{|1\rangle}(t) = \frac{\Omega^2}{\Omega^2 + (\omega - \omega_0)^2} \sin^2\left(\frac{\sqrt{\Omega^2 + (\omega - \omega_0)^2}}{2} t\right)$$

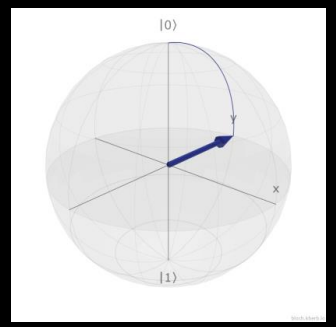
- Off-resonance ($\omega \neq \omega_0$):
  - The maximum probability to reaching $|1\rangle$ is $\frac{\Omega^2}{\Omega^2+(\omega-\omega_0)^2} < 1$.
  - The qubit never fully reaches $|1\rangle$.
- On-resonance ($\omega = \omega_0$): $P_{|1\rangle}(t) = \sin^2\left(\frac{\Omega}{2} t\right)$
  - The $B_1 \cos(\omega t)$ field can drive the qubit between $|0\rangle$ and $|1\rangle$!
  - These are called Rabi oscillations.

Controlling Qubits

$$P_{|1\rangle}(t) = \sin^2\left(\frac{\Omega}{2}t\right)$$

- We can control the qubit by applying the drive $B_1 \cos(\omega_0 t)$ for particular durations.
  - π pulse:
    - $P_{|1\rangle}\left(t = \frac{\pi}{\Omega}\right) = 1$
    - $|0\rangle \rightarrow |1\rangle$
    - Complete flip
  - $\frac{\pi}{2}$ pulse:
    - $P_{|1\rangle}\left(t = \frac{\pi}{2\Omega}\right) = \frac{1}{2}$
    - $|0\rangle \rightarrow \frac{|0\rangle - i|1\rangle}{\sqrt{2}}$
    - Beamsplitter



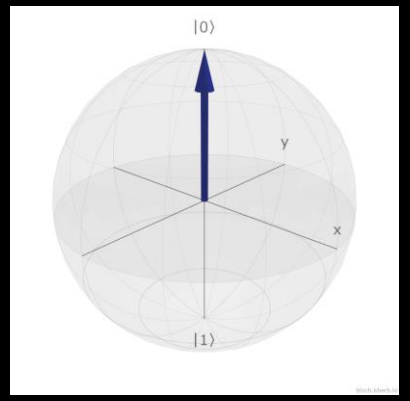


Ramsey Sequence

Phase
$\phi = \delta\omega\, T$
Detuning
$\delta\omega = \omega - \omega_0$

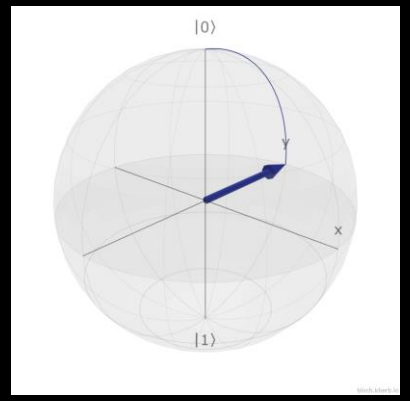
Initialize

$|0\rangle$



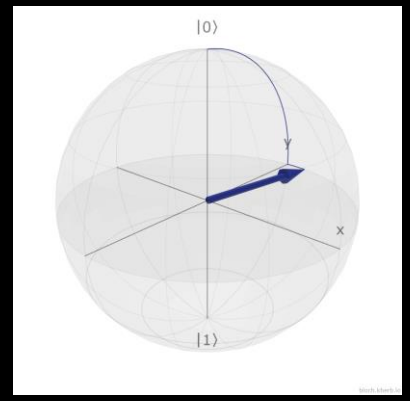
Apply $\pi/2$ pulse

$$\frac{|0\rangle - i|1\rangle}{\sqrt{2}}$$



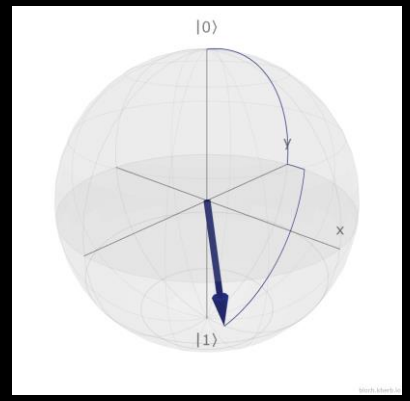
Free evolution for time T

$$\frac{|0\rangle - ie^{-i\phi}|1\rangle}{\sqrt{2}}$$



Apply $\pi/2$ pulse

$$\sin\left(\frac{\phi}{2}\right)|0\rangle - i\cos\left(\frac{\phi}{2}\right)|1\rangle$$



Measurement along z

$$P_{|0\rangle} = \sin^2\left(\frac{\phi}{2}\right)$$

$$P_{|1\rangle} = \cos^2\left(\frac{\phi}{2}\right)$$

Precision: $\delta\omega \sim \frac{1}{T}$

Application to Quantum Sensing
Quantum Clocks

- Quantum clocks have lowest systematic uncertainty of any device!
  - If operated for the age of the universe, they won't lose a second.
- Any regularly oscillating system can function as a clock by counting the number of cycles.
  - More rapid oscillations = more sensitive clock
- Atomic clock: Lock a laser's frequency to an atomic transition, driving oscillations using Ramsey sequence
  - No B_0 field: natural two level system
  - B_1 pulses: laser pulses

PHYSICAL REVIEW LETTERS **133,** 023401 (2024)

Editors' Suggestion | Featured in Physics

Clock with 8×10^{-19} Systematic Uncertainty

Alexander Aeppli,[1,*] Kyungtae Kim,[1] William Warfield,[1] Marianna S. Safronova,[2] and Jun Ye[1,†]

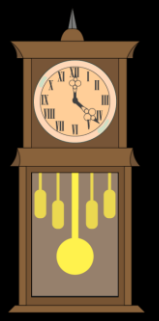
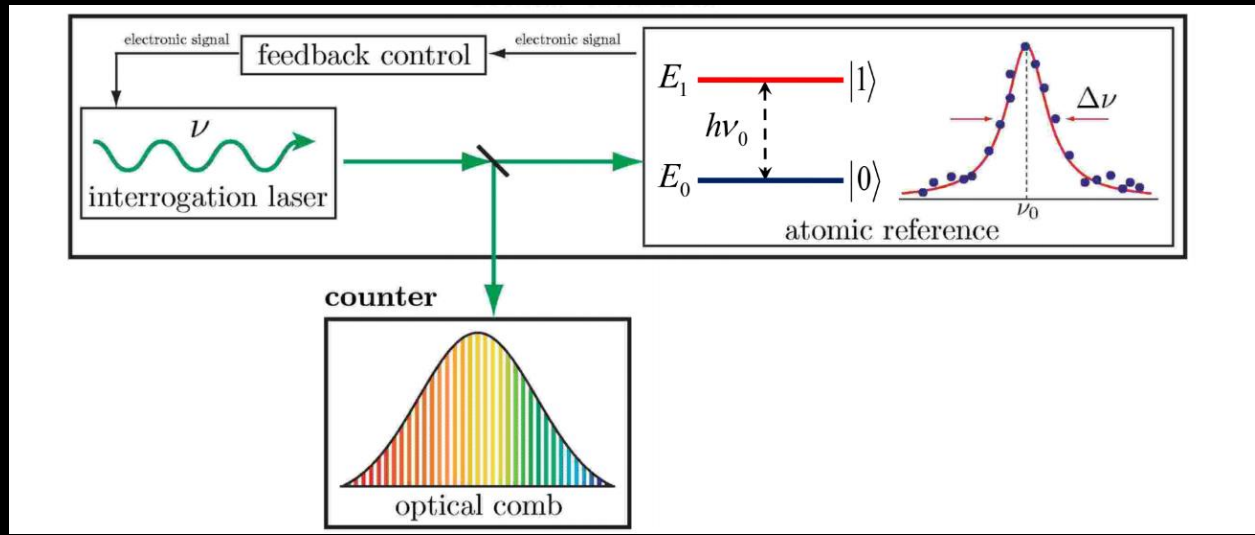
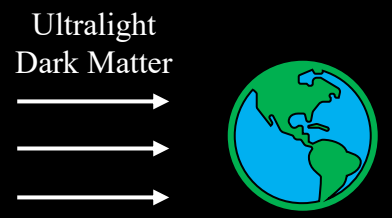


Figure Credit: N. Poli, et. al. *Riv. Nuovo Cimento.* 36, 12, 555 (2013)



Application to Quantum Sensing Dark Matter Detection

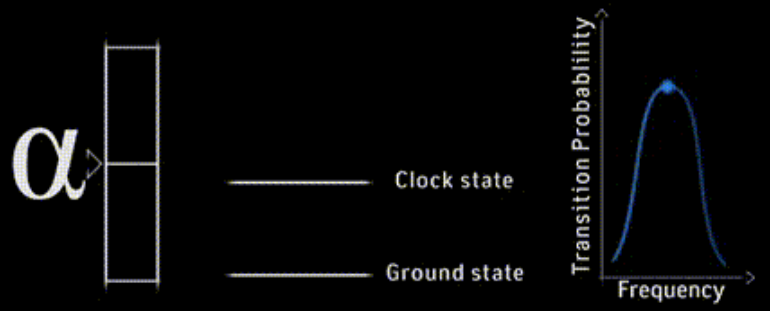


- Ultralight scalar dark matter is a class of wave-like dark matter models that induce variations in fundamental constants (such as α).

$$\alpha \rightarrow \alpha + \delta\alpha$$

- Atomic transition frequencies depend on these fundamental constants.

$$\nu \propto \alpha^K$$



- The scalar dark matter can thus cause variation in atomic frequencies.

$$\nu \rightarrow \nu + \delta\nu$$

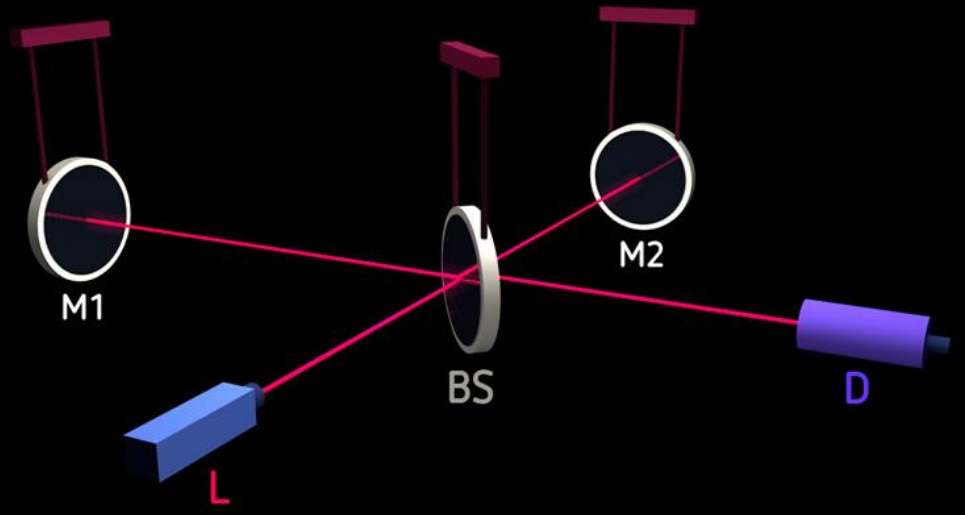
$$\frac{\delta\nu}{\nu} = \frac{K\alpha^{K-1}\,\delta\alpha}{\alpha^K} = K\frac{\delta\alpha}{\alpha}$$

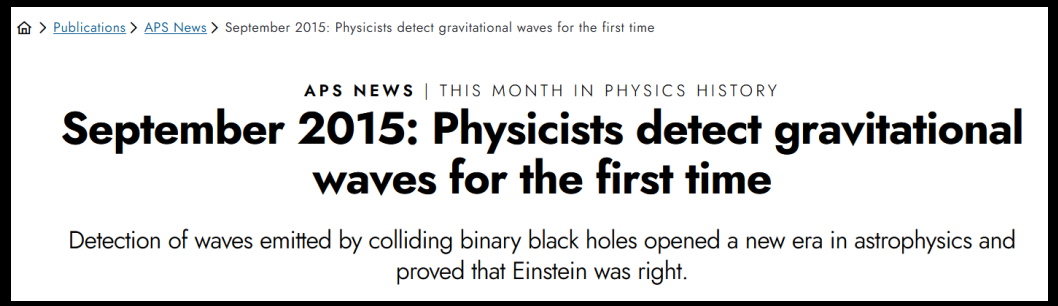
- Ramsey sequence in quantum clocks detects $\delta\nu$ – dark matter signal appears as oscillating shift in atomic transition frequency.

Figure Credit: https://www.learner.org/series/physics-for-the-21st-century/the-quantum-world/

Application to Quantum Sensing: Gravitational Wave Detection

- $|0\rangle, |1\rangle$ are the two paths
- $\frac{\pi}{2}$ pulse is the beam splitter BS
- Free precession is light L traveling down arms, accumulating phase
- Phase accumulated is due to difference in path length between arms
  - This is caused by a passing gravitational wave stretching one arm and compressing the other
- Second $\frac{\pi}{2}$ pulse is beam splitter BS on returning light from mirrors M1, M2
- Measurement: intensity of light at detector D




Publications > APS News > September 2015: Physicists detect gravitational waves for the first time

APS NEWS | THIS MONTH IN PHYSICS HISTORY

September 2015: Physicists detect gravitational waves for the first time

Detection of waves emitted by colliding binary black holes opened a new era in astrophysics and proved that Einstein was right.

$\Delta L \sim 10^{-18}$ m – thousandth of the diameter of a proton

https://www.einstein-online.info/en/spotlight/Interferometric-gravitational-wave-detectors/

Summary

- Spin ½ particle in magnetic fields:
  - Static field $B_0 \rightarrow$ Larmor precession at $\omega_0 = \gamma B_0$
  - Resonant drive $B_1 \cos(\omega t) \rightarrow \frac{\pi}{2}$ and π pulses
  - Ramsey sequence:
    - Superposition + phase accumulation + readout → frequency sensor with precision $\delta\omega \sim \frac{1}{T}$

